

HIGH-VOLTAGE TESLA COIL SYSTEM

Solid-State Resonant Step-Up Design Powered by a ZVS Driver & Flyback Stage

Project Domain: High-Voltage Physics & Power Electronics **Topology:** ZVS Mazzilli Oscillator + Flyback Driver

Input Power: 12V - 24V DC (High Current)

Secondary Frequency: ~250 kHz - 500 kHz (RF)

1. Executive Summary & Objectives

This project report documents the design, construction, mathematical analysis, and performance validation of a compact **Solid-State Tesla Coil (SSTC) System**. The primary objective is to build an efficient, robust high-voltage high-frequency AC generator by combining a self-oscillating **Zero Voltage Switching (ZVS) Mazzilli Driver**, a center-tapped **Flyback Transformer (FBT)**, and a resonant **Tesla Secondary Coil**.

Key Project Objectives:

- High Efficiency Driver Implementation:** Construct a ZVS oscillator operating with soft switching to minimize switching losses ($P_{sw} \approx 0$) and limit thermal dissipation across MOSFET switches.
- Intermediate High-Voltage Generation:** Utilize a ferrite-core flyback transformer to step up a low DC voltage (12V–24V) to a high-frequency AC output ranging from 10kV to 20kV.
- Resonant Energy Transfer:** Achieved resonance matching between the driver frequency and the LC natural frequency of the secondary coil to produce high-potential plasma arcs and streamers.
- Wireless Power Transmission Demonstration:** Demonstrate electromagnetic near-field induction by illuminating gas-discharge tubes (fluorescent and neon lamps) wirelessly without physical contact.

2. System Architecture & Block Diagram

The system operates in three distinct stages: Power Conditioning & ZVS Inversion, Flyback Step-Up, and Resonant Step-Up via the Tesla Coil Secondary.



Functional Flow Explanation: Low-voltage DC electrical energy enters the self-oscillating ZVS driver, which converts DC into high-frequency AC (typically 30 kHz – 80 kHz). This AC signal drives the center-tapped primary of the flyback transformer. The step-up output of the flyback transformer feeds into the primary stage of the Tesla coil system, which resonates with the high-Q secondary winding to produce multi-kilovolt high-frequency electrical fields at the terminal top load (toroid).

3. Components Used & Technical Specifications

The table below provides a complete Bill of Materials (BOM) along with the functional specifications for each component used in the system build:

Component Name	Specification / Value	Quantity	Functional Role in Circuit
Power MOSFETs	IRFP250N / IRFZ44N (N-Channel)	2 Pcs	Main power switches operating in push-pull arrangement for the ZVS oscillator.
Resonant Tank Capacitor	0.68 μ F / 0.33 μ F 630V MKP/FKP	1 - 2 Pcs	Forms the LC resonant tank circuit alongside the primary inductor winding.
Toroidal Choke Inductors	100 μ H - 200 μ H (High Current Wire)	2 Pcs	Supplies steady DC current to MOSFET drains and decouples switching noise.
Fast Switching Diodes	UF4007 / 1N4148	2 Pcs	Provides cross-coupled gate discharge paths to ensure rapid MOSFET turn-off.
Zener Diodes	12V, 1W Power Rating	2 Pcs	Clamps gate-source voltage (V_{GS}) to protect gates from overvoltage.
Gate Pull-Up Resistors	470 Ω to 1 k Ω , 2W	2 Pcs	Provides turn-on bias current to the MOSFET gates from the DC supply.
Gate Pull-Down Resistors	10 k Ω , 0.25W	2 Pcs	Ensures clean grounding of gates during idle state to prevent false triggering.
Flyback Transformer (FBT)	Line Output Transformer (TV FBT)	1 Pc	High-frequency ferrite-core step-up transformer with custom 5+5 primary winding.
Tesla Secondary Coil	32-36 AWG Enameled Copper Wire	1 Coil (~800-1000 turns)	High-Q inductor winding on PVC pipe for high-frequency voltage multiplication.
Top Load (Toroid)	Aluminum Duct / Smooth Metallic Sphere	1 Pc	Provides isotropic capacitance (C_{top}) to store energy prior to discharge.

4. Working Principle & Circuit Dynamics

4.1 ZVS Mazzilli Driver Operation

The ZVS driver circuit is an ultra-efficient self-oscillating push-pull converter based on the Royer topology. Its operation relies on zero-voltage switching, which dramatically reduces power losses during semiconductor transitions.

Zero Voltage Switching Principle: Switching transitions only occur when the voltage across the active switching MOSFET drops to zero ($V_{DS} = 0V$). This eliminates overlap between high voltage and high current during turn-on/turn-off, preventing heavy thermal dissipation.

Step-by-Step Circuit Sequence:

- Startup Asymmetry:** When 12V DC power is applied, current flows through the 470Ω resistors to both MOSFET gates. Due to minor manufacturing tolerances, one MOSFET (e.g., Q1) turns on slightly faster than the other (Q2).
- Conduction & LC Tank Excitation:** As Q1 conducts, its drain voltage drops to near ground ($V_{DS1} \approx 0V$). The fast diode D1 grounds the gate of Q2, keeping Q2 firmly turned off. Current now flows through one half of the center-tapped flyback primary winding into the LC tank capacitor.
- Resonant Sinusoidal Swing:** The resonant tank circuit formed by the inductor (L_p) and capacitor (C_l) creates a sinusoidal voltage waveform across the primary winding. As the tank voltage swings back across zero, the voltage at Q1's drain rises while Q2's drain voltage falls towards 0V.
- Self-Toggled Switching:** Once Q2's drain voltage drops near zero, diode D2 turns off and allows Q2's gate to charge via its resistor, turning Q2 ON. Simultaneously, diode D1 pulls Q1's gate to ground, turning Q1 OFF. This process repeats indefinitely at the resonant frequency:

$$f_{ZVS} = \frac{1}{2\pi\sqrt{L_p \cdot C_l}}$$

4.2 High-Voltage Flyback Step-Up Stage

The high-frequency AC oscillation generated by the ZVS driver is injected into the 5+5 turn center-tapped primary wound on the flyback transformer's high-permeability ferrite core. Due to the high frequency operation (30–60 kHz), the ferrite core operates near saturation without overheating, stepping up the voltage proportionally to the turns ratio:

$$V_{out} = V_{in} \times (N_{secondary} / N_{primary})$$

This generates an intermediate high-voltage output of approximately **10,000V to 20,000V AC** at high frequency from the flyback secondary terminals.

4.3 Tesla Resonant Coupling & Air Breakdown

The high-voltage output from the flyback stage feeds the primary stage of the Tesla coil. The Tesla coil acts as a resonant transformer operating on double resonance. When the primary driving frequency aligns perfectly with the secondary's natural resonant frequency:

$$f_1 = \frac{1}{2\pi\sqrt{L_1 C_1}} = f_2 = \frac{1}{2\pi\sqrt{L_2 C_2}}$$

Energy continuously transfers between the primary and secondary inductors over multiple cycles. Due to the high Q-factor of the secondary coil, the voltage at the toroid top load scales to extremely high levels, exceeding the dielectric breakdown threshold of air (~30 kV/cm). This leads to localized ionization and intense plasma arc discharges (corona discharges and purple streamers).

5. Experimental Results & Performance Observations

During testing and validation of the constructed Tesla coil setup, the following electrical and physical operational characteristics were measured and verified:

- **Operating Supply Voltage:** 12V DC nominal (tested up to 18V DC under regulated cooling).
- **Current Consumption:** 4.5A – 7.2A (depending on load distance and arc length).
- **Output Arc Length:** Continuous plasma arcs of 3.5 cm – 6.0 cm produced into grounded metallic probes.

- **Wireless Power Radius:** A standard 18W fluorescent tube illuminated brightly when brought within 25 cm - 35 cm of the top load without any wired connections.
- **Thermal Performance:** MOSFET heatsinks remained at moderate temperatures ($\sim 42^{\circ}\text{C}$) during continuous 3-minute runs, confirming effective Zero Voltage Switching action.

6. Safety Considerations & Precautions

CRITICAL HIGH VOLTAGE SAFETY WARNING:

- **High Voltage Hazard:** The output of the flyback transformer and Tesla coil secondary exceeds several kilovolts and can cause severe electric shock or burns. Always maintain a safe operating distance.
- **RF Energy & Thermal Burns:** Although high-frequency RF current exhibits the "skin effect," high-power RF can still cause deep tissue thermal burns. Never touch the discharge arc directly.
- **Electromagnetic Interference (EMI):** High-frequency electrical discharges generate broadband RF noise. Sensitive electronics, smartphones, and microcontrollers should be kept at a distance or shielded.
- **Ozone & UV Safety:** Spark discharges produce trace amounts of ozone and UV radiation. Ensure tests are conducted in a well-ventilated area.

7. Conclusion & Future Enhancements

The construction of the **ZVS Driver and Flyback-Driven Tesla Coil** was successfully completed and validated. The project demonstrated the principles of resonant energy transfer, power electronics switching topologies, and high-voltage physics. The Zero Voltage Switching topology proved highly efficient, keeping switching heat to a minimum while delivering strong excitation power to the high-voltage flyback transformer and secondary resonant coil.

Future Scope & Upgrades:

- **Musical / Audio Modulation:** Integrate a pulse-width modulation (PWM) or microcontroller-based interrupter stage into the gate drive lines to modulate the spark frequency and play musical tones via plasma discharge.
- **Dual Resonant Solid-State Tesla Coil (DRSSTC) Conversion:** Upgrade the primary stage to a full primary resonant tank with IGBT switching to achieve multi-foot streamer lengths.
- **Enclosure & Active Cooling:** Design a custom 3D-printed/acrylic safety enclosure with dual cooling fans for prolonged continuous demonstration.